

Diagnosing Quadrant and Parameter-Orientation Errors

Answer Key

Precalculus

Quick Answers

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| 1. (a) the corrected angle = $\frac{3\pi}{4}$, — | 6. (a) the x-coordinate = -2 , (a) the |
| 2. (a) the corrected angle = $\frac{4\pi}{3}$, — | y-coordinate = $-2\sqrt{3}$, — |
| 3. (a) the x-coordinate = 0 , (a) the | 7. (a) the angles where r is zero = $\frac{\pi}{4}$, $\frac{3\pi}{4}$, $\frac{5\pi}{4}$, |
| y-coordinate = -2 , — | $\frac{7\pi}{4}$, — |
| 4. (a) x at $t = -2 = -1$, (a) x at $t = 1 = 2$, — | 8. (a) x at $t = 0 = 5$, (a) y at $t = 0 = -1$, — |
| 5. (a) y when $x = 2 = 1$, — | |
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What is verified

12 of 20 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-IF.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

- If they got -0.79 : kept the raw calculator arctangent instead of adding π for a second-quadrant point
 - If they got 1.05 : reported the raw arctangent, which lands in the first quadrant instead of the third
 - If they got 2 : used the size of r and dropped its minus sign, landing in the wrong quadrant
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1. A student converts $(-3, 3)$ to polar form, enters $\arctan\left(\frac{3}{-3}\right)$ and writes $\theta \approx -0.79$. (a) Correct θ in $[0, 2\pi)$. (b) Name the error.

Solution (a): $\arctan\left(\frac{3}{-3}\right) = \arctan(-1) = -\frac{\pi}{4} \approx -0.79$. The point has $x < 0$, so

the true direction is half a turn away: $-\frac{\pi}{4} + \pi$.

$\theta = \frac{3\pi}{4}$

Solution (b) — instructor-judged. The arctangent function returns values only in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so it can never report a second-quadrant direction. With x negative and y positive

the point lies in the second quadrant, and π must be added to the calculator's answer. Full credit needs both the range limitation and the quadrant of the point.

2. A student converts $(-1, -\sqrt{3})$ the same way and reports $\theta \approx 1.05$. (a) Correct θ . (b) Name the error and the role of the two negative signs.

Solution (a): $\frac{-\sqrt{3}}{-1} = \sqrt{3}$, and $\arctan(\sqrt{3}) = \frac{\pi}{3} \approx 1.05$. Since $x < 0$, add π : $\frac{\pi}{3} + \pi$.

$$\theta = \frac{4\pi}{3}$$

Solution (b) — instructor-judged. Both coordinates are negative, so the point is in the third quadrant. The two minus signs cancel inside the quotient y/x , so the ratio looks exactly like a first-quadrant point's ratio and the arctangent returns a first-quadrant angle. The sign information has to be restored by adding π . Full credit requires naming the cancellation as the cause.

3. A student says $x = 2 \cos t$, $y = -2 \sin t$ traces $x^2 + y^2 = 4$ counterclockwise from $(2, 0)$. (a) Find the point at $t = \pi/2$. (b) Name the error.

Solution (a): $x = 2 \cos \frac{\pi}{2} = 2(0)$, so

$$x = 0$$

and $y = -2 \sin \frac{\pi}{2} = -2(1)$, so

$$y = -2$$

Solution (b) — instructor-judged. At $t = 0$ the curve is at $(2, 0)$ and a quarter turn later it is at $(0, -2)$ — it has moved *down*, so the motion is clockwise. The rectangular equation $x^2 + y^2 = 4$ is correct but carries no orientation; the minus sign on the sine is what reverses the direction. Full credit needs the clockwise conclusion supported by the two sampled points.

4. A student writes: for $x = t + 1$, $y = t^2$, the curve is traced right to left as t increases. (a) Find x at $t = -2$ and $t = 1$. (b) Name the error.

Solution (a): At $t = -2$: $x = -2 + 1$, so

$$x = -1$$

At $t = 1$: $x = 1 + 1$, so

$$x = 2$$

Solution (b) — instructor-judged. $x = t + 1$ is increasing in t , and the two samples confirm it: x moves from -1 up to 2 as t runs from -2 to 1 . The curve is therefore traced left to right, so the claim reverses the orientation. Full credit for any argument that shows x increasing with t .

5. For $x = \sqrt{t}$, $y = t - 3$, $t \geq 0$, a student gets $y = x^2 - 3$ and draws the entire parabola. (a) Find y when $x = 2$. (b) Name the error.

Solution (a): $y = 2^2 - 3 = 4 - 3$.

$$y = 1$$

Solution (b) — instructor-judged. The algebra is right but the domain is not. $x = \sqrt{t}$ is never negative, so the parametric curve consists only of the points with $x \geq 0$ — the right half of the parabola. The entire left branch the student drew is not on the curve. Full credit requires the restriction $x \geq 0$ and the identification of the left half as the extra piece.

6. A student converts $(-4, \frac{\pi}{3})$ to rectangular form as $x = 2$, first quadrant. (a) Find the correct coordinates. (b) Name the error.

Solution (a): Substitute the signed r : $x = (-4) \cos \frac{\pi}{3} = (-4) \left(\frac{1}{2}\right)$, so $x = -2$

and $y = (-4) \sin \frac{\pi}{3} = (-4) \left(\frac{\sqrt{3}}{2}\right)$, so $y = -2\sqrt{3}$

Solution (b) — instructor-judged. The student used $|r| = 4$ and dropped the minus sign. A negative first coordinate reverses the ray, so the point lies half a turn from the $\pi/3$ direction — in the *third* quadrant, at $(-2, -2\sqrt{3})$, not the first. Full credit needs the dropped sign named and the third quadrant identified.

7. A student claims $r = 4 \cos(2\theta)$ never has $r < 0$. (a) Find every θ in $[0, 2\pi)$ with $r = 0$. (b) Explain what happens between the first two of them.

Solution (a): $4 \cos(2\theta) = 0$ when $2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$, i.e.

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Solution (b) — instructor-judged. Between $\theta = \pi/4$ and $\theta = 3\pi/4$ the cosine of 2θ is negative, so r is negative there. Those points are plotted half a turn from the direction the angle names, which is why the petal generated by second-quadrant angles actually appears in the fourth quadrant. That reversal is exactly how a four-petal rose fills all four quadrants. Full credit requires "r is negative on that interval" plus the half-turn relocation.

8. For $x = 2 + 3 \cos t$, $y = -1 + 3 \sin t$, a student writes $(x - 2)^2 + (y + 1)^2 = 3$. (a) Find the point at $t = 0$. (b) Show it fails and correct the equation.

Solution (a): $x = 2 + 3 \cos 0 = 2 + 3$, so $x = 5$

and $y = -1 + 3 \sin 0 = -1 + 0$, so $y = -1$

Solution (b) — instructor-judged. Substituting $(5, -1)$ into the student's equation gives $(5 - 2)^2 + (-1 + 1)^2 = 9$, not 3 — the point on the curve does not satisfy it. The identity $\cos^2 t + \sin^2 t = 1$ gives $(x - 2)^2 + (y + 1)^2 = 3^2$, so the right-hand side must be the square of the radius:

$$(x - 2)^2 + (y + 1)^2 = 9$$

Full credit needs the failed substitution and the corrected right-hand side.